

Durability and degradation mechanisms of GFRP reinforcement subjected to severe environments and sustained stress

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ABSTRACT

Despite the large amount of research that has been carried out to date on the use of glass fibre reinforced polymer (GFRP) bars in concrete, one factor still hindering their widespread use in civil engineering applications is the lack of comprehensive data on their long-term in-service performance. This paper presents the test results of an experimental study investigating the physical and mechanical properties of GFRP bars exposed to severe environments and subjected to different levels of sustained load. The test environments included moist concrete, alkaline solution and tap water, with temperatures varying from 20 °C to 60 °C. The mechanical properties of the bars were characterized through direct tension, flexural and inter-laminar shear tests, while the physical and chemical properties were determined through the implementation of a series of complementary techniques, including moisture absorption measurements, scanning electron microscope (SEM), Fourier Transform Infrared Spectroscopy (FTIR) and Energy dispersive X-ray analyses (EDX). The test results showed that the elevated temperatures play a key role in triggering and accelerating the development of critical degradation mechanisms. The reduction in the tensile strength of all conditioned samples subjected to a sustained stress equivalent to 3000 $\mu\epsilon$ was always within the limits recommended in existing codes for high durability bars, while a lower average strength retention was observed for higher levels of sustained stress (equivalent to 5000 $\mu\epsilon$). Finally, it can be concluded that the long-term mechanical properties of the tested GFRP bars appeared to be mainly affected by moisture diffusion through the resin rich layer and debonding at the fibre/matrix interfaces due to the dissolution of the silane coupling agents.

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1. Introduction

Corrosion of steel reinforcement is one of the main causes of degradation of reinforced concrete (RC) structures. In order to reduce the high maintenance cost associated with this problem, alternative reinforcing materials, such as Fibre Reinforced Polymer (FRP) reinforcements, have been investigated over the past three decades [1–4]. Despite the large amount of research that has been carried out on the use of FRP in concrete, the lack of comprehensive data on their long-term in-service performance [3,5] still hinders the widespread use of FRP bars in civil engineering applications.

The durability of Glass FRP (GFRP) bars in concrete has usually been evaluated by using accelerated test methods that expose the bars to environments harsher than those they may encounter in service and by studying the change with time of their mechanical

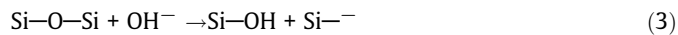
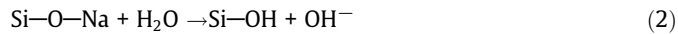
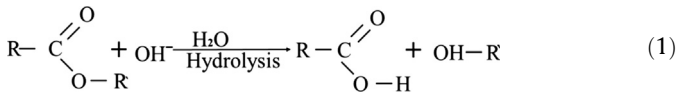
properties (i.e. tensile strength, bond strength and modulus of elasticity) [2,6–10]. In an attempt to achieve a more fundamental understanding of their long-term behaviour, several researchers also investigated changes in physical, chemical and microstructural properties using scanning electron microscopy (SEM) [4,11], energy dispersive X-ray analyses (EDX) [11,12], dynamic mechanical analysis (DMA) [13], thermogravimetric analysis (TGA) [13,14], differential scanning calorimetry (DSC) [10,14] and Fourier transform infrared spectroscopy (FTIR) [10,15].

Among the conditioning environments examined in the literature, moist and alkaline environments have been reported to be the most detrimental for the GFRP matrix and the fibres [6,16–19]. In particular, in moist environments the free hydroxide ions (OH^-) attack the polymer matrix and the long molecular chains are disassembled by hydrolysis, promoting the ingress of more OH^- ions and water molecules (H_2O) (Eq. (1)). These, in turn, can break the polymer-polymer chain secondary bond, creating localized voids (increase in free volume) and ultimately affecting the

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physical and mechanical properties of the GFRP matrix (i.e. plasticization) [6]. The presence of water can also lead to the degradation of glass fibres through dissolution (i.e. leaching) (Eq. (2)). This process, which consists in the extraction of alkalis from the glass structures, continues until alkalis are available. The hydroxide produced by the leaching process (Eq. (2)) increases the pH of the solution and, as the pH exceeds the threshold value of 9, the Si networks are affected (i.e., silica dissolution in alkaline environment, Eq. (3)) [19].



While direct submersion of bare bars in an alkaline and water solution at elevated temperature is recommended by current standards to accelerate aging [20–22], previous studies [11,18,19] have shown that these environments are excessively harsh and cause premature deterioration of GFRP bars, consequently lead to conservative predictions of service life. As a result, researchers attempted to evaluate the performance of GFRP bars by embedding them directly into concrete and exposing the specimens to high temperatures to accelerate reaction rates, yet subjecting them only to the inherent alkaline and moist environment provided by the concrete surrounding the bars [9,18,23].

In addition to the exposure environment, the presence of sustained stress might generate stress concentrations around the micro-imperfections present in the matrix (e.g. pores) and accelerate the propagation of microcracks [24]. This, in turn, would ease the ingress of the surrounding media (e.g. alkali solution and water) and have a possible detrimental impact on the durability of the fibres [25]. Only a limited number of studies examined the influence of sustained stress on the long-term properties of FRP and concluded that low values of sustained stress (20% of the ultimate strength or less) did not significantly affect the residual mechanical properties. Such studies, however, only examined the performance of bare composite specimens [26], or were limited to relatively low values of sustained stress (equivalent to a strain

level of about 2000 $\mu\epsilon$) and did not include any chemical/physical analysis of the microstructure [27].

2. Research significance

This study aims to provide a better understanding of the long-term performance of GFRP bars by employing accelerated conditioning methodologies that more closely reflect the in-service conditions of GFRP RC members including alkaline and moist environments as well as sustained stresses. A comprehensive experimental programme was designed to study the degradation mechanisms at both micro (chemical properties) and macro levels (physical and mechanical properties). Ultimately, the goal of this research is to develop an effective model to assess and predict the durability of GFRP bars, which in turn, would directly assist in the development of more reliable and less conservative design tools for GFRP RC members.

3. Experimental programme

Fig. 1 presents a flow chart summarizing the complementary experimental program carried out in this research. The flow chart is divided in three sections: Section 1 – Degradation; Section 2 – Investigation; and Section 3 – Model. Section 1 shows the conditioning parameters (i.e. moist and alkali environments, high temperature and sustained stress) considered in this research and the GFRP bar properties that are affected (at both the macro level – tensile and bond properties – as well as at the micro level). Section 2 summarizes the tests used to assess the investigated material properties. The use of this complementary set of tests was implemented to enable a more comprehensive assessment of the degradation processes based not only on the outcomes of the mechanical tests but also on the results of the physical and chemical tests. Section 3 is presented in the chart as the logical convergence of this study, however a durability model is still being developed and therefore will not be discussed in this paper.

3.1. Test matrix

The test matrix Table 1 consisted of 334 GFRP samples obtained from bars with a nominal diameter of 8 mm. The GFRP bars used in

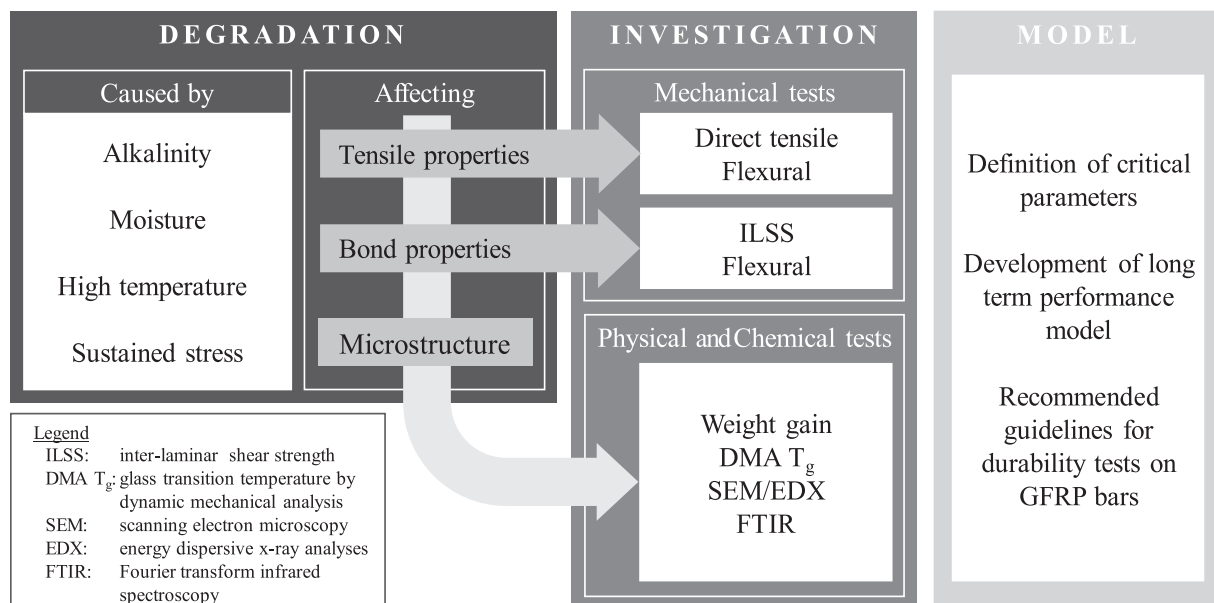


Fig. 1. Overview of experimental program.

Table 1
Test matrix.

Conditioning			Test method						
Exposures	Temp	Time	MAb	TNS	ILSS	FLX	DMA	FTIR	SEM/EDX
	(°C)	(hr)	(N°)	(N°)	(N°)	(N°)	(N°)	(N°)	(N°)
REF	20	0		5	5	5	3	5	5
K	20	1000, 2000, 6480, 8760	3	9	3	3	1	3	2
	40		3	9	3	3	1	3	2
	60		3	9	3	3	1	3	2
K-S3	20	1000, 2000, 6480, 8760	4	9	3	3	N.A.	3	2
	40		4	9	3	3		3	2
	60		4	9	3	3		3	2
W	20	1000, 2000, 6480, 8760	3	N.A.	3	3	1	3	2
	40		3		3	3	1	3	2
	60		3		3	3	1	3	2
W-S3	20	1000, 2000, 6480, 8760	4	N.A.	3	3	N.A.	3	2
	40		4		3	3		3	2
	60		4		3	3		3	2
CON	20	2000, 6480, 8760	N.A.	9	N.A.	N.A.	N.A.	N.A.	N.A.
M-CON	20	1000, 2000, 6480	N.A.	9	N.A.	N.A.	N.A.	N.A.	N.A.
	40			9					
	60			9					
M-CON-S3	20	1000, 2000, 6480	N.A.	9	N.A.	N.A.	N.A.	N.A.	N.A.
	40			9					
	60			9					
M-CON-S5	60	42, 90, 270	N.A.	9	N.A.	N.A.	N.A.	N.A.	N.A.

Note: TNS, tensile test; MAb, water absorption test; ILSS, inter-laminar shear strength; FLX, flexural tests; DMA, T_g by dynamic mechanical analysis; FTIR, Fourier transform infrared spectroscopy; SEM, scanning electron microscopy; EDX, energy dispersive X-ray analyses; REF, reference samples; K, alkaline solution; W, tap water; M, moisture; CON, concrete; S3 and S5 sustained stress at 3000 $\mu\epsilon$ and 5000 $\mu\epsilon$, respectively.

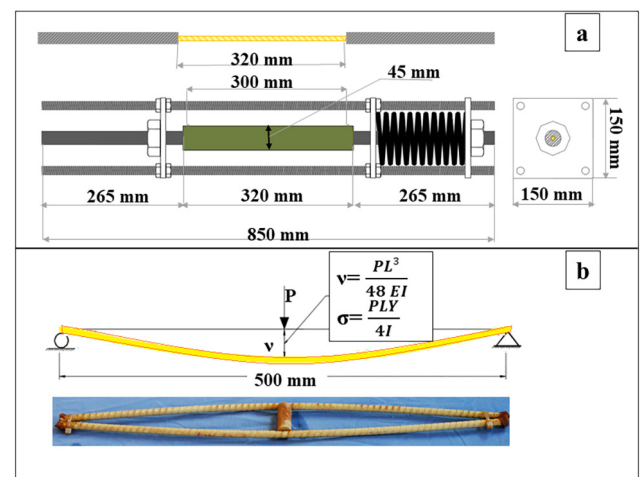
this research work are commercially available and made of continuous E-CR Glass fibres impregnated with a vinyl ester resin and a glass fibre content of approximately 88% by weight. After the bars have gone through the pultrusion process, ribs are cut in the hardened bars and their surface is subsequently coated (Fig. 2).

The program included tests on: unconditioned reference specimens (REF); unstressed and stressed bare bars conditioned in alkaline solution (K and K-S3) or tap water (W and W-S3) at 20, 40 and 60 °C; unstressed and stressed bars embedded in concrete (i.e. moist and alkali environment) at ambient temperature (CON) and immersed in tap water (M-CON, M-CON-S3 and M-CON-S5) at 20, 40 and 60 °C. The exposure durations taken into account varied from 1000 hr (~42 days) to 8760 hr (1 year).

Although it has been shown in previous research that direct exposure of GFRP to alkaline solution is undoubtedly more aggressive than any realistic civil engineering application, such condition has been included as it is recommended in current testing guidelines [21,22]. The alkaline solution was prepared according to the recommendations of the ACI 440 committee [22] using 118.5 g of $\text{Ca}(\text{OH})_2$, 0.9 g of NaOH and 4.2 g of KOH in 1 L of deionised water. The pH level of this solution was periodically controlled and kept at about 12.7, which is a representative value for a mature concrete pore solution [28].

**Fig. 2.** Appearance and surface geometry of the bar specimens.

A sustained stress inducing a tensile strain of 3000 $\mu\epsilon$ was applied to the FRP bars as recommended in ACI 440.3R and CAN/CSA-S806 [22,21], while a higher stress inducing a strain of 5000 $\mu\epsilon$ was examined to assess the effect of less stringent serviceability limits (corresponding to larger allowable crack widths and deflections) [29]. The sustained stress inducing the desired level of tensile strain was applied using two different configurations: via a spring of adequate stiffness mounted in a stiff pre-tensioning rig (Fig. 2a); or by tying pairs of specimens at the two ends with a wedge interposed at mid-length to impose the required curvature (Fig. 3b). During the conditioning period, the strain level of the bars was measured periodically using a caliper and demec gauge system. The length of the steel springs used in the loading system

**Fig. 3.** Sketch illustrating the methods of application of sustained stress for (a) tensile and (b) moisture absorption test specimens.

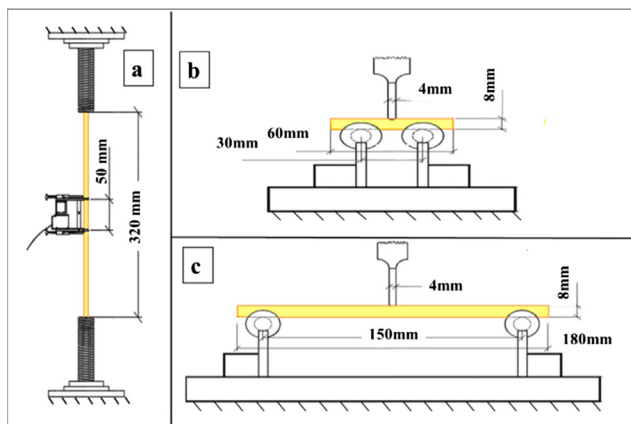


Fig. 4. Set up for (a) tensile, (b) inter-laminar shear strength and (c) flexural tests.

was also checked regularly using a digital caliper with an accuracy of 0.02 mm to assure that the load was maintained constant.

3.2. Test methodology

Direct tensile tests (TNS) [22], inter-laminar shear strength tests (ILSS) [30] and flexural tests (FLX) [31] were carried out to evaluate the mechanical properties. All tests were performed in displacement control using hydraulic and mechanical actuators with a maximum capacity of 1000 kN (TNS) and 10 kN (ILSS and FLX), respectively. Data were recorded at 1 Hz using an NI LabVIEW data acquisition system. Details of the setup of each test are shown in Fig. 4.

A series of tests on the physical and chemical properties of the samples was also carried out to investigate the microstructure and the integrity of the GFRP material before and after exposure. Standard test methods were implemented to measure the glass transition temperature (T_g) by dynamic mechanical analysis (DMA) [32] (Fig. 5a) as well as moisture absorption (MAb) [33]. DMA was performed using a Mettler Toledo DMA/SDTA861e machine while moisture absorption was assessed monthly by measuring the weight gain due to fluid uptake into 300-mm-long (unstressed) and 450-mm-long (stressed) GFRP samples with the two ends sealed using epoxy to prevent absorption from the cut edges. Prior to conditioning, the specimens were dried in an oven at 55 °C for 24 h. The moisture uptake was monitored monthly using a digital scale with accuracy of 0.001 g. For each set of measurements, the specimens were removed from their respective conditioning environments at the same time, their surface was wiped dry with a cloth, and weighed immediately. In addition, scanning electron microscopy (SEM), energy dispersive X-ray analyses (EDX) and Fourier transform infrared spectroscopy analysis (FTIR) were

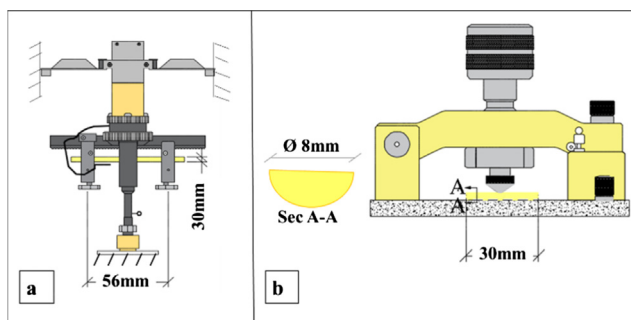


Fig. 5. Set up for (a) T_g and (b) FTIR tests.

carried out following procedures available in the literature [12,34] (Fig. 5b).

SEM and EDX were performed using a high magnification microscope (Phillips XL30 SEM) which uses a focussed scanned electron beam to produce images of the sample. Prior to the analysis, 10-mm-long samples were cut from the GFRP bars using a low speed diamond blade. The samples were subsequently placed vertically in moulds, cold-mounted in epoxy resin and cured for 24 h at room temperature. The cross sections were then ground using sand paper of increasing grit (400, 600, 800 and 1200) and polished with 6 μ m and 1 μ m diamond pastes. The specimens were then carbon coated and analysed. Finally, FTIR analyses were performed using an FTIR spectrometer to investigate the external surface and the internal layers of the GFRP bars. In the former case, 30-mm-long samples were cut longitudinally from the bars up to a maximum depth of 3 mm, while in the latter the samples were prepared by grinding the GFRP bars into a fine powder (approximately 2 mg). The milled samples were pressed into a disc after being mixed with approximately 200 mg of anhydrous Potassium Bromide (KBr). The disc analysis was carried out by treating the KBr as the background reference. In both configurations, 32 scans were routinely obtained with an optical retardation of 0.25 cm to yield a resolution of 4 cm^{-1} . Typical characteristic absorption bands available in the literature are reported in Table 2. Past studies on FRP examining civil engineering applications specifically focused on spectral zones in the band between 3600 and 2900 cm^{-1} including the stretching mode of the hydroxyl group OH (3450 cm^{-1}) and of the carbon-hydrogen group CH (2928 cm^{-1}) [12,34]. In this research the acquisition of the spectra was performed ranging from 4000 to 600 cm^{-1} , so as to include the stretching modes of C–O and C=C (1295 cm^{-1} and 920 cm^{-1} respectively).

4. Results and discussion

The results of the experimental programme are presented in this section. Firstly, the results from the moisture absorption tests, which can be used to investigate changes in the diffusion mechanism of water and alkaline solutions, are reported and used to support the results of physical and chemical tests, which assessed the deterioration of the micro structure (i.e., matrix, fibres and interface layer). Finally, the data acquired from all tests is utilized to discuss the changes in mechanical properties at the macro scale.

4.1. Moisture absorption properties

Fig. 6 shows the absorption rate of stressed (S3) and unstressed GFRP bars in water (W) and alkaline (K) environment against the square root of the exposure time. The absorption rate was calculated as the ratio between the average moisture uptake and the initial weight.

In all samples, weight increased gradually with respect to the exposure time, following the Fick's law up to about 4500 h

Table 2
Assignments of the main absorption bands of the FTIR spectra.

Absorption bands (cm^{-1})	Assignment
3400	O–H bending vibration
3026	C–H stretch of the benzene
2960, 2928, 2871	CH, CH ₂ , CH ₃ stretch
1722	C=O— Stretching vibration in saturated aldehyde, ketone or acid
1608, 1510	C=C— Stretching vibration of skeleton in benzene ring
1295	C–O stretch
1160	C–CO–C stretch
920	C=C stretch

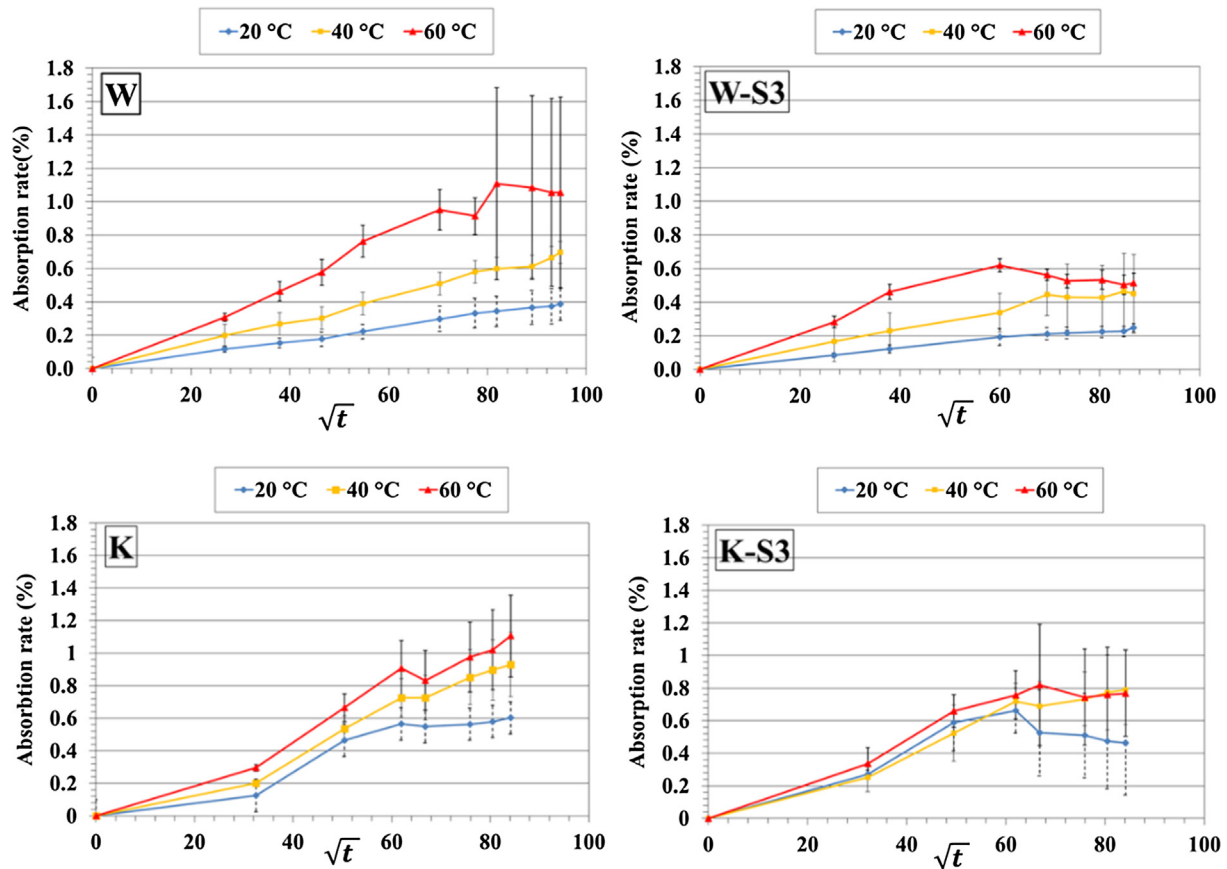


Fig. 6. Moisture absorption rate of unstressed and stressed (S3) GFRP bars immersed in water (W) and alkaline solution (K) at different temperature.

($\sim 67 \sqrt{\text{hr}}$). At this stage, the diffusion mechanism started to become unstable (non-Fickian behaviour) resulting in absorption rate curves characterized by portions of weight loss, due to the degradation of the outer layer leading to material loss, and by portions of weight gain, possibly due to the filling of new or pre-existing voids and microcracks. While no saturation was reached at the end of the considered ageing period (8760 h–93 $\sqrt{\text{h}}$), stressed samples consistently showed a lower moisture uptake than unstressed ones. This phenomenon could be attributed to the method used to stress the samples (i.e. bending, Fig. 3b) which reduces the number of open pores in the portion of the bar in compression. This is discussed further in Section 4.2.2. In addition, the exposure temperature played a critical role on the absorption process. The measured data showed that the higher the temperature the higher the absorption rate. This can be attributed to an increase in the chemical reactions rate as well as to a higher void pressure resulting from an increase in the volume of gases as temperature rises. This pressure build-up favoured the propagation of microcracks and consequently increased the free volumes within the bars that could be filled by the surrounding solution.

It can be finally observed that, for all temperatures, specimens exposed to alkali solution showed slightly higher moisture uptake than those exposed to tap water as the former contains more free hydroxide ions (OH^-), which work as a solvent for the polymer, thus promoting the solution ingress.

4.2. Chemical and physical degradation of GFRP bars in wet environments

4.2.1. Effects on matrix

The effect of the different conditioning environments, temperatures and applied stress levels, on the GFRP matrix was studied at

different depths of the samples. In particular, the external surface was analysed using both SEM images and FTIR. FTIR was also employed to examine an inner portion of the bars, yet still sufficiently close to the surface, while the core of the samples was analysed using DMA.

FTIR was employed to detect possible chemical changes in the polymeric chain after conditioning. Intensity changes in the hydroxyl group (OH) (Eq. (1)) and in the stretching bands of C=O and C–O were investigated to verify that hydrolysis and oxidation reactions took place. Since the intensity of a FTIR spectrum can be affected by several parameters, it is customary [19] to quantify the changes in a given band as a ratio with the variations of the aromatic group CH, which is very stable and does not easily break apart and react with other substances. Thus, changes in the OH/CH (I3400/I3026) and C=O/CH (I1722/I3026) ratios as well as in the C–O/CH (I1295/I3026) ratio can be used to gage hydrolysis and oxidation reactions, respectively. Similarly, changes in the C=C peak, also referred to as curing index [35], were monitored to gain information on the curing process.

The FTIR spectra were calculated on the surface of control (REF), unstressed (W and K) and stressed (W-S3 and K-S3) samples (Fig. 7a) as well as on a deeper portion of control (REF) and unstressed (W and K) samples (Fig. 7b, ground samples). All samples were conditioned for 8760 h at 60 °C.

The results of the FTIR analysis on the surface of the samples (Table 3) showed that all the OH/CH ratios increased with respect to REF and that such increase was more marked in unstressed samples than in stressed ones. This directly confirms that moisture entered the polymer matrix promoting the hydrolysis reaction and it also implies that stressed samples had lower chemical interaction (i.e., lower hydrolysis). As the intensity of OH was similar for unstressed samples (only slightly lower for K), the C=O/CH (I1722/

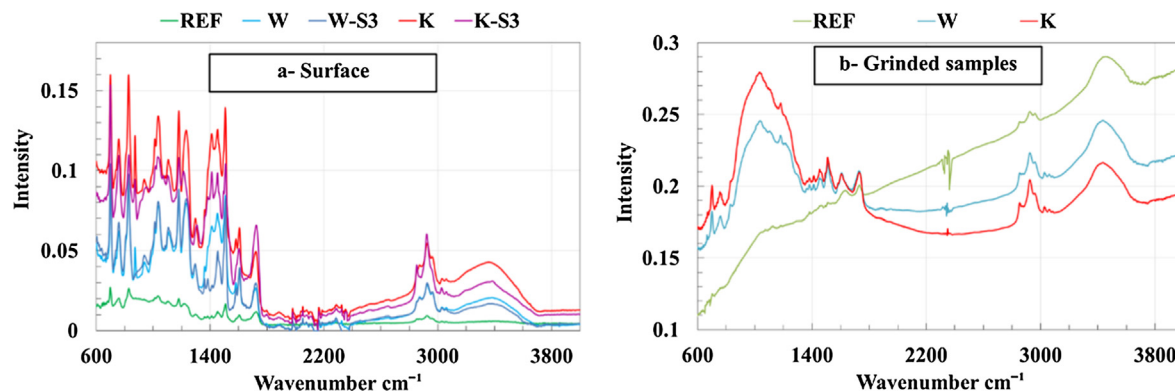


Fig. 7. FTIR spectra for the resin matrix on (a) the surface of GFRP bars and on (b) ground samples. Results for unstressed and stressed (S3) specimens conditioned in water (W) and alkali solution (K) at 60 °C for 8760 h are compared to benchmark samples (REF).

Table 3

Band ratios results for FTIR analysis on the outer surface of the samples aged at 60 °C for 8760 h.

Samples	OH/C—H (I3400/I3026) (1)		C=O/C—H (I1722/I3026) (2)		C—O/C—H (I1295/I3026) (3)		C=C/C—H (I920/I3026) (4)		C=C/C—H (I1608/I3026) (5)	
	Value	(%)	Value	(%)	Value	(%)	Value	(%)	Value	(%)
REF	1.05	—	2.07	—	1.79	—	1.49	—	3.45	—
W	1.40	1.33	1.82	0.88	2.80	1.56	2.47	1.66	2.92	0.85
W-S3	1.16	1.10	2.10	1.01	2.94	1.64	2.73	1.83	3.32	0.96
K	1.32	1.26	1.52	0.73	2.16	1.21	1.98	1.33	2.89	0.84
K-S3	1.16	1.10	2.46	1.19	2.46	1.37	1.87	1.26	3.36	0.97

I3026) ratio was used as an additional indicator of the hydrolysis reaction taking place and to assess the effect of the conditioning environment. The reduction in C=O/CH (I1722/I3026) was due to the breakage of the end of the vinyl ester chain and, as suggested by the lower value, the alkaline solution seemed to cause more breakage of the ester chain than water (i.e. more hydrolysis). Conversely, while oxidation occurred in all samples as all the C—O/CH (I1295/I3026) ratios increased with respect to REF, this reaction

was more evident in stressed samples than unstressed ones and in water rather than alkaline solution.

Table 4 shows that, for all ground deeper samples, there was a small reduction of the OH/CH ratio and a negligible variation of the C=O/CH (I1722/I3026) ratio with respect to the REF samples, suggesting that no hydrolysis reaction occurred in this portion of the bar. The test results also showed an increase in the ratio of C—O/CH (I1295/I3026) ratio in both K and W samples due to the

Table 4

Band ratios results for FTIR analysis on a ground portion of the inner layer of the samples aged at 60 °C for 8760 h.

Samples	OH/C—H (I3400/I3026) (1)		C=O/C—H (I1722/I3026) (2)		C—O/C—H (I1295/I3026) (3)		C=C/C—H (I920/I3026) (4)		C=C/C—H (I1608/I3026) (5)	
	Value	(%)	Value	(%)	Value	(%)	Value	(%)	Value	(%)
REF	1.17	—	0.57	—	0.70	—	0.79	—	0.62	—
W	1.13	0.97	0.57	1.00	1.00	1.43	0.98	1.24	1.08	1.74
K	1.13	0.97	0.57	1.00	1.19	1.70	1.11	1.41	1.41	2.27

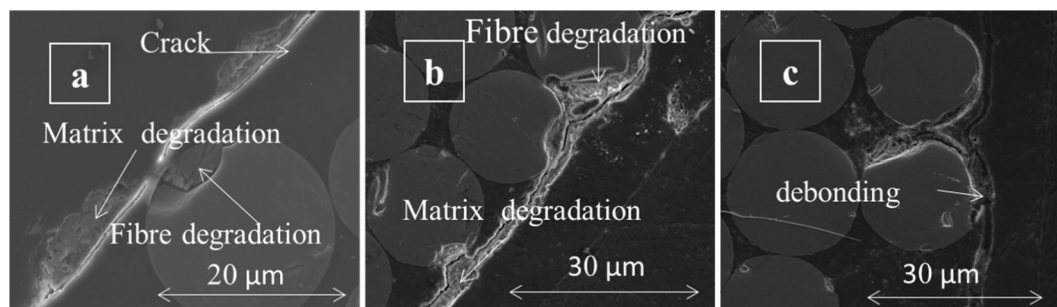


Fig. 8. SEM images of the cross-sections of specimen conditioned in (a) water and (b and c) alkaline solution at 60 °C for 8760 h.

oxidation of the methylene group (CH_2) at the end of the vinyl ester chain.

For both surface and ground specimens, an increase in the $\text{C}=\text{C}/\text{CH}$ (I920/I3026) ratio with respect to REF was observed (column 4 in Tables 3 and 4). This may be attributed to additional repeating units connecting to the backbone chain and increasing the number of benzene rings, and consequently the stiffness of the resin. The ratio of $\text{C}=\text{C}/\text{CH}$ at different locations of the polymer chain (i.e. I608/I3026) (column 5 in Tables 3 and 4) showed a reduction in the outer layer and an increase in the inner part of the bars, thus providing evidence of degradation on the surface and post-curing of the core portion of the bars.

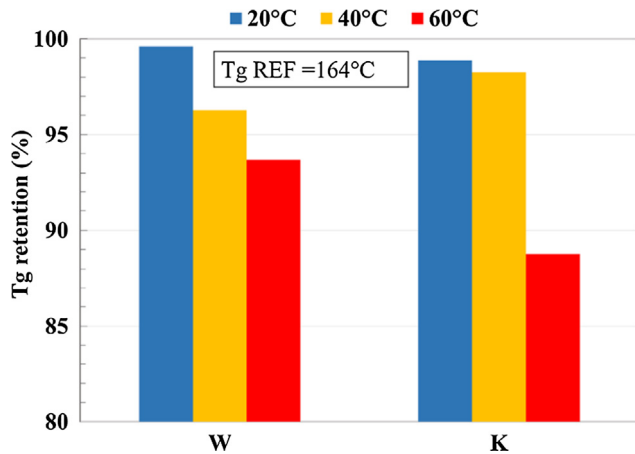


Fig. 9. Change in glass transition temperature (T_g) for the core of specimens conditioned in water (W) and alkaline solution (K) at different temperatures for 8760 h.

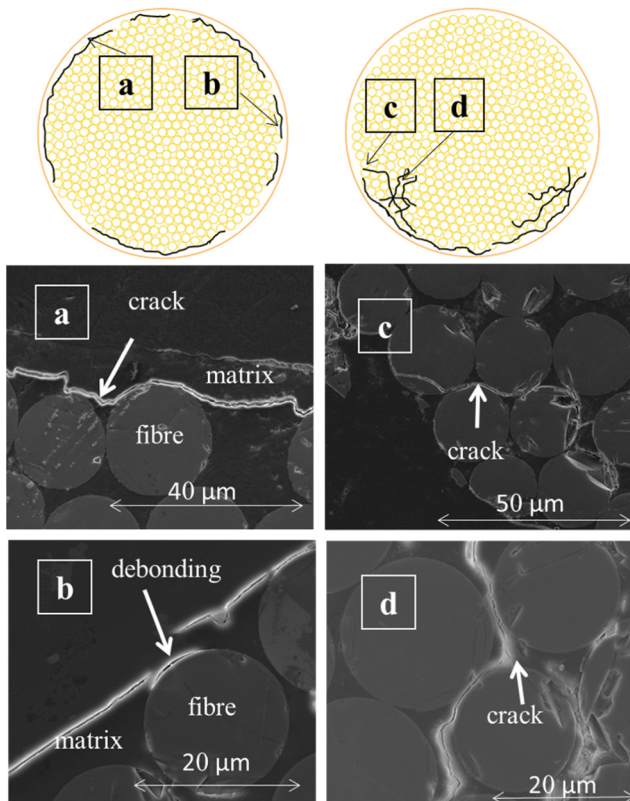


Fig. 10. SEM images of the cross sections of unstressed specimen (left) and stressed specimens (right) conditioned in water at 60 °C for 8760 h.

Additional evidence of matrix deterioration (i.e. cracks, voids and de-bonding from the fibres) was provided by the SEM images of the outer layers of GFRP bars conditioned in water (W) and alkaline solution (K) at 60 °C for 8760 hr (Fig. 8).

The analysis of the innermost part of the GFRP bars was carried out by determining the glass transition temperature (T_g) of samples obtained from the core of specimens aged for 8760 h in water (W) and alkaline solution (K) at 20 °C, 40 °C and 60 °C. After the specimens were removed from the conditioning environment, they were stored in sealed plastic bags up to the time of testing. The values of T_g were determined according to the storage modulus (E') method. Fig. 9 shows that the change in T_g was negligible for low temperatures, while a reduction of more than 5% was observed at 60 °C for both environments. Since a reduction in T_g indicates a reduction of transfer properties of resin matrix, it can be concluded that, at least at elevated temperatures, the matrix properties in the core of the specimens were also affected.

All the results confirm that in presence of moisture the higher the conditioning temperatures the higher the deterioration of the resin matrix and indicate that the polymer chain undergoes disruption (hydrolysis) and reconstruction (oxidation) of the bond links with possible repercussion on the physical and mechanical properties of the polymer matrix.

4.2.2. Effects on matrix/fibre interface

The interface between the matrix and the fibres is a very thin layer (approximately 1 μm) of coupling agent, consisting of a saline-based material susceptible to moisture and alkaline environments. The assessment of the deterioration of such a thin layer was only possible by using highly magnified SEM images.

Fig. 10 shows, at the top, the typical crack pattern observed in the cross-section of unstressed (left) and stressed (right) samples, as well as the approximate locations where the SEM images (a–d) were taken. As no significant deterioration was seen in samples conditioned at low temperatures, only images of samples conditioned in water and alkaline solution at 60 °C for 8760 h are presented. These results confirm that moisture dissolved the saline substances at the interface causing local loss of bond between fibres and matrix (i.e., cracks and debonding). In addition, it can be noticed that for the unstressed sample the cracks tend to prop-

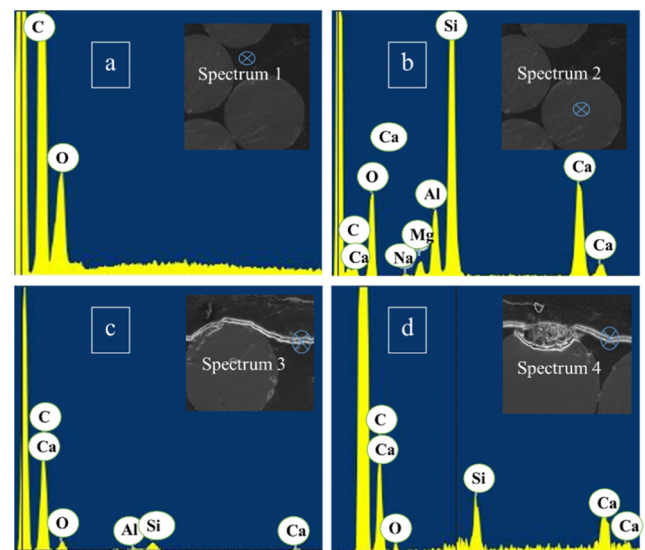


Fig. 11. Results of EDX scans for specimens conditioned (c) in water and (d) alkaline solution at 60 °C for 8760 h are compared to benchmark samples for (a) matrix and (b) fibres.

agate circumferentially around the outermost layer of fibres, while the cracks are concentrated on the tension side of the stressed samples and propagate both radially and circumferentially as a result of the three-point bending setup used to stress the specimens (Fig. 3b).

4.2.3. Effects on glass fibres

Highly magnified SEM images were used to study the degradation of the glass fibres in water and alkaline solution (Eqs. (2) and (3)). In addition, as silica (Si), aluminium (Al) and calcium (Ca) are not chemical elements present in a vinylester matrix, a precise elemental analysis using EDX was carried out to detect the presence of such elements as a result of the glass fibres dissolution.

The results of the EDX analysis carried out on the matrix and the fibres of a reference sample (Fig. 11a and b, respectively) and on the matrix of samples conditioned in water and alkaline solution at 60 °C for 8760 h (Fig. 11c and d, respectively) are presented in the form of the spectra displaying the concentration (peaks) of each chemical element identified in the analysis.

In all figures, the location where the analysis was performed is indicated by a target. The EDX analysis detected the aforementioned chemical elements in the resin matrix near to the fibre/matrix interface both for specimens conditioned in water (W, Fig. 11c) and alkaline solution (K, Fig. 11d). This confirmed that water molecules and hydroxide ions, through micro-cracks in the matrix, reached the glass fibres deteriorating them. Additional evidence

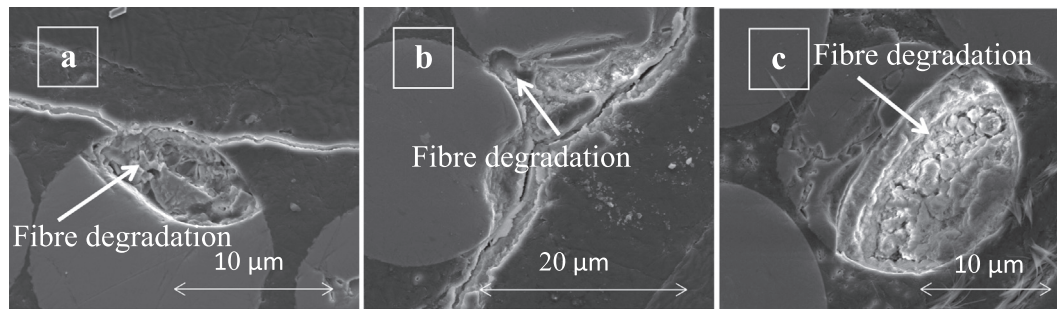


Fig. 12. SEM images showing the degradation in glass fibres after conditioning in alkaline solution at 60°C for 8760 h.

Table 5
Tensile test results.

Conditioning			Tensile strength			Modulus of elasticity		
Exposures	Temp (°C)	Time (hr)	Avg (MPa)	COV (%)	Retention (%)	Avg (GPa)	COV (%)	Retention (%)
REF	20	0	1542	1.8	100	56.1	2.7	100
K	20	1000	1472	0.3	95	56.7	0.7	101
	20	2000	1454	0.2	94	53.6	0.6	96
	20	6480	1498	n.a.*	97	58.5	n.a.*	104
	40	1000	1407	1.9	91	57.5	4.0	102
	40	2000	1390	1.3	90	57.0	4.5	102
	40	6480	1310	n.a.*	84	60.4	n.a.*	108
	60	1000	1310	3.1	84	59.2	6.2	106
	60	2000	1139	11.6	73	58.0	1.7	103
	60	6480	997	n.a.*	64	66.4	n.a.*	118
K-S3	60	1000	1327	5.4	86	58.9	4.4	105
	60	2000	1164	4.5	75	56.1	4.2	100
CON	20	6480	1540	0.7	100	56.7	6.4	101
M-CON	20	2000	1421	5.3	92	55.3	3.4	99
	20	6480	1392	6.9	90	55.5	3.5	99
	40	1000	1458	1.4	95	56.5	0.1	101
	40	2000	1370	1.0	89	57.5	1.9	102
	40	6480	1148	18.0	74	58.4	5.4	104
	60	1000	1244	5.9	80	59.8	6.0	107
	60	2000	1227	1.1	79	58.8	4.2	105
	60	6480	917	1.4	59	55.4	1.9	99
M-CON-S3	20	1000	1473	0.2	96	57.6	1.0	103
	20	2000	1416	1.8	92	61.6	6.5	110
	20	6480	1365	11.5	89	55.5	3.0	99
	40	1000	1228	17.0	87	58.5	0.3	104
	40	2000	1335	3.6	87	57.0	4.2	102
	40	6480	1177	3.6	76	57.2	2.7	102
	60	1000	1233	2.4	82	58.3	2.8	104
	60	2000	1073	11.0	74	58.8	2.7	105
	60	6480	919	4.2	59	61.1	3.4	109
M-CON-S5	60	1000	1190	14.0	77	59.4	2.2	106
	60	2000	1023	0.6	66	59.9	0.9	107
	60	6480	971	4.8	63	58.3	5.0	104

* COV not available since only one specimen was tested.

of the fibre deterioration within the outer layer was provided by SEM images as shown in Fig. 12.

4.3. Mechanical degradation of GFRP bars in wet environments under sustained stress

4.3.1. Tensile properties

The results of the tensile tests are summarized in Table 5, including average (Avg) tensile strength and modulus of elasticity, their coefficients of variation (COV) and their retention calculated as percentage of results for REF. The tensile strength was calculated using a nominal diameter of 8 mm while the modulus of elasticity was calculated as the slope of the stress-strain curve between 20% and 60% of the ultimate tensile strength.

Fig. 13 shows the tensile strength and moisture absorption rate over the conditioning time for unstressed and stressed specimens exposed to moist concrete (M-CON, M-CON-S3, and M-CON-S5) and alkaline solution (K and K-S3) at 20, 40 and 60 °C. The results indicate that tensile strength decreased as moisture uptake increased and these effects were accelerated by increasing the temperature. It should be mentioned that moisture absorption and strength retention were measured on different specimens configurations (see Fig. 3) and the different stress distribution within the direct tension specimens might have affected moisture absorption, which was possibly underestimated as a result of the different induced cracking pattern.

The reduction in tensile properties of GFRP bars can be interpreted in light of the results obtained by FTIR and SEM/EDX. The deterioration of the matrix allowed the ingress of hydroxide ions with a twofold effect: (1) causing de-bonding between fibres and matrix, consequently reducing the GFRP ability to transfer effectively the load from the matrix to the fibres; (2) dissolving the glass fibres, thus resulting in a local reduction of the number of fibres effectively contributing to carry the load.

In addition, even if the test results showed that the changes in the average tensile modulus of elasticity of most conditioned specimens are negligible, as it varied within the variability of the experimental data, a general increasing trend can be noticed. This beneficial effect on the stiffness was particularly evident in specimens directly exposed to alkaline solution (K) for an extended period of time (6480 h), showing that the higher the temperature the higher the modulus of elasticity. This could be interpreted as post-curing of the matrix, phenomenon that was also previously captured by FTIR analysis in the inner layers of the bar. In addition, and as supported by the analysis of the SEM micrographs, only a small portion of the bar was subjected to a significant level of deterioration after exposure to the conditioning environments. The majority of the tensile strength reduction occurred within the first 2000 h for all environments, with a slightly higher degradation rate being observed for the specimens conditioned in alkaline solution (K). The unstressed specimens embedded in concrete (M-CON), exhibited a slightly higher retention rate up to 2000 h but their strength at 6480 h was overall lower than that of specimens immersed in alkaline solution. This can be attributed to the formation of salt barriers on the surface of the K specimens that inhibited further diffusion and damage to the microstructure. The application of a sustained stress during conditioning, equivalent to 3000 $\mu\epsilon$ and 5000 $\mu\epsilon$, affected slightly only the tensile strength, possibly because the level of imposed strain was small compared to the ultimate strain of the REF samples (i.e. 27000 $\mu\epsilon$).

4.3.2. Flexural properties and inter-laminar shear strength

Table 6. summarizes the results of flexural (flexural strength and modulus of elasticity) and inter-laminar shear tests (ILSS) in terms of average values, coefficients of variation and retention for unstressed and stressed bars directly exposed to water (W

and W-S3) and alkaline solution (K and K-S3) at 20, 40 and 60 °C for 6480 h.

The retention of mechanical properties is also shown in Figs. 14 and 15 along with the moisture absorption (MAB) rate results to illustrate the effects of moisture ingress on flexural and inter-laminar properties.

It can be noticed that the average flexural strength of all conditioned specimens decreased compared to the reference specimens. In particular, for exposures at 20 and 60 °C, the higher the temperature, the lower the flexural strength, which is in line with the findings previously discussed for TNS and MAB tests. Conversely, at 40 °C, the application of sustained stress appears to affect strength retention significantly, as the strength of stressed samples was much lower than that of unstressed ones. However, this behaviour was only recorded for flexural tests and the same trend was

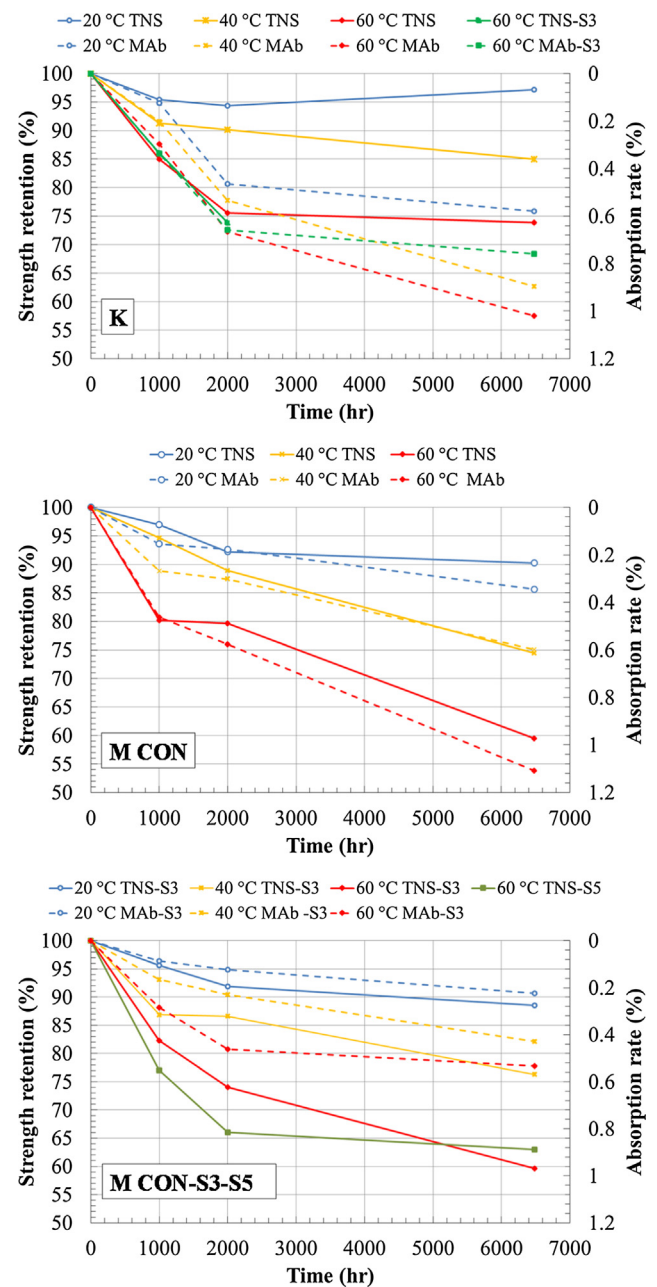
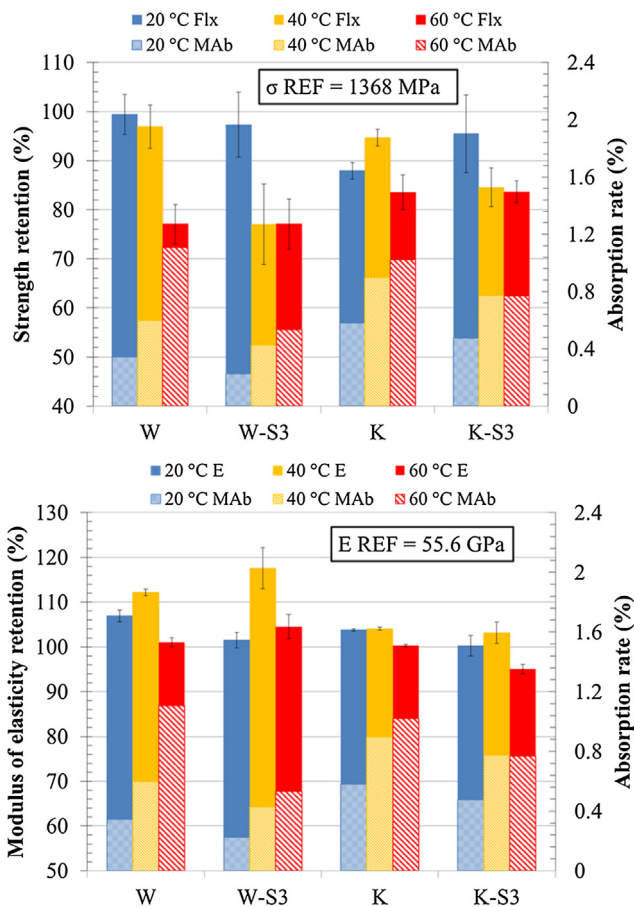


Fig. 13. Tensile strength (TNS) retention and moisture absorption (MAB) over time for unstressed and stressed (S3 and S5) bars conditioned in alkaline solution (K) and concrete (M-CON).

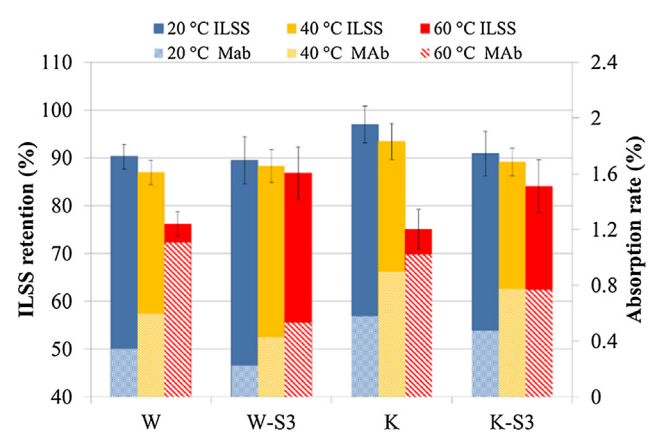
Table 6

The results of flexural (flexural strength and modulus of elasticity) and inter-laminar shear tests (ILSS).

Conditioning		Flexural Strength			Modulus of Elasticity			ILSS		
Exp.	T (°C)	Avg (MPa)	COV (%)	Retention (%)	Avg (GPa)	COV (%)	Retention (%)	Avg (MPa)	COV (%)	Retention (%)
REF	20	1368	7.1	100	55.6	1.5	100	73	2.5	100
W	20	1360	4.0	99	59.5	1.3	107	66	4.8	90
	40	1326	4.3	97	62.4	0.7	112	63	3.3	86
	60	1124	4.0	82	56.2	1.0	101	55	5.4	75
W-S3	20	1331	6.6	97	56.1	1.7	101	65	3.8	89
	40	1054	8.2	77	65.4	4.6	115	64	3.7	87
	60	1054	5.0	77	58.2	2.7	104	63	4.0	86
K	20	1203	1.6	88	57.8	0.2	103	70	1.6	95
	40	1296	3.5	94	57.9	0.3	103	68	3.5	93
	60	1143	4.0	83	55.8	0.2	100	54	4.0	74
K-S3	20	1306	7.9	95	55.8	2.2	100	66	6.2	90
	40	1157	3.9	84	57.4	2.3	103	64	1.7	87
	60	1145	2.2	83	52.92	2.2	95	61	1.3	85

**Fig. 14.** Flexural strength (Flx) and modulus of elasticity (E) are compared to moisture absorption results (MAB) for different conditioning environments after 6480 h.

not observed in TNS and ILSS tests results. This can be attributed to the different state of stress and failure mode generated within specimens subjected to different tests configurations. Although flexural tests are relatively simple to perform, the different behaviour of composite materials in tension and compression (with unidirectional FRP being typically characterised by a lower compressive strength) can result in underestimating tensile strength and in an increased variability of results. The variability in the results obtained from flexural tests can also increase for conditioned specimens, especially when deterioration occurs at the

**Fig. 15.** Interlaminar shear strength retention (ILSS) is compared to moisture absorption results (MAB) for different conditioning environments after 6480 h.

fibre/matrix interface and can affect compression behaviour. As stiffness is always determined within the perfectly elastic load-deformation range, the values determined for flexural stiffness generally agree well with those obtained from direct tension tests.

The average values of the flexural modulus of elasticity increased marginally for most of the specimens, with the exception of samples conditioned in water at 40 °C for which the flexural stiffness increased more than 10%. The overall higher stiffness could have been caused by post-curing of the matrix during conditioning. This is in line with the results obtained from the direct tension tests as well as the FTIR findings, albeit at lower temperature.

The results from the ILSS tests show that the higher the conditioning temperature, the higher the strength degradation, but also that stressed specimens experienced lower reduction in ILSS than unstressed ones. Such findings are in good agreement with previous tests carried out in this study, suggesting that the increased temperature favoured the ingress of moisture leading to a premature inter-laminar de-bonding and that the state of stress induced using the test setup described in Fig. 3b limited moisture ingress mitigating the reduction in ILSS.

5. Conclusions

The present study investigated the mechanical performance and physical and chemical characteristics of GFRP bars conditioned in different environments with and without sustained

stress. The mechanical and microstructural properties were evaluated by direct tension test, flexural tests, inter-laminar shear tests, moisture absorption, SEM-EDX, FTIR, and DMA. On the basis of the discussion presented above, the following conclusions can be drawn:

- 1) Moisture uptake of the tested GFRP bars occurred at a faster rate up to 4600 h. The different rate of diffusion can be attributed to the moisture absorption via capillaries and micro-cracks in the resin matrix. This mechanism continues until moisture reaches the fibres, which provide a barrier and restrict moisture flow. The tests results also confirmed that diffusion rate during the initial stages is highly dependent on the temperature, and higher temperatures lead to higher absorption rates. However, the specimens exposed to alkaline solution showed slightly higher moisture uptake than those exposed to tap water as the former contains more free hydroxide ions (OH^-), which work as a solvent for the polymer, thus promoting ingress of the solution.
- 2) For temperature up to 40 °C the degradation observed in specimens exposed to K and M-CON environments were similar, whereas for higher temperatures and after 2000 h of exposure less degradation was observed in specimens exposed to K environment. This can be attributed to deposition of salts on the bars surface, which provided a barrier and restricted moisture diffusion into the bars, thus potentially leading to underestimate strength retention.
- 3) Unexpectedly, the stressed specimens exhibited lower absorption rate than the unstressed counterparts. This may be attributed to the method that was used to induce the desired strain (i.e. imposing the required curvature via elastic bending) and the closing of microcracks within the compression zone of the specimens. Further tests on bars subjected to different stress distributions (e.g. uniform tension) should be carried out to examine this aspect in more details.
- 4) The tensile test results clearly show that the tensile strength of the tested GFRP bars was affected by the conditioning environments. In general, higher temperatures and higher levels of sustained stress lead to higher strength degradation. Similar trends were observed for all types of exposure with most of the tensile strength reduction occurring within the first 2000 h. This can be attributed to the high diffusion of water molecules through the resin matrix outer layer inducing damage to the fibre/resin interface. After 2000 h exposure, bars embedded in concrete under sustained stress equivalent to 5000 $\mu\epsilon$ conditioned in water at 60 °C exhibited the higher reduction in strength due to the acceleration of moisture uptake caused by the high temperature and as a result of the higher level of damage in the matrix caused by the sustained stress.
- 5) The reduction in the tensile strength of all conditioned samples subjected to a sustained stress equivalent to 3000 $\mu\epsilon$ was within the limits recommended in the Canadian code for high durability bars [21]. A lower average strength retention was observed only for the specimens subjected to the higher sustained stress equivalent to 5000 $\mu\epsilon$.
- 6) No significant change was observed in the elastic modulus of the tested GFRP bars regardless of the conditioning environment or exposure period. Although a small degree of degradation was observed through SEM and EDX in the outer resin matrix layer and some of the glass fibres closer to the surface of the bar, the observed level of damage was too small to result into an appreciable stiffness reduction. Some of the samples exhibited a slight enhancement in stiffness, possibly due to the post curing of the resin matrix, as confirmed by FTIR results.

- 7) The change in flexural properties (e.g. strength, stiffness) was in line with what observed as a result of direct tension tests.
- 8) No significant fibre deterioration was observed through SEM/EDX within the cross-section of the bars, with the majority of the damage being limited to a small percentage of glass fibres within the outer surface layer. Microstructural changes in the matrix and debonding between fibre/matrix seem to be the main issues affecting the long-term mechanical properties of the tested GFRP bars.
- 9) The EDX scan detected additional elements such as silicon (Si), aluminium (Al), and calcium (Ca) in the interface zone and in the resin matrix. This is attributable to the dissolution of the fibres and leaching of their components into the resin matrix.
- 10) The FTIR analysis showed a significant increase in the amount of hydroxyl groups at the surface of the material. This is an indication of chemical degradation in the resin matrix.

In summary, the long-term mechanical properties of GFRP bars appear to be mainly affected by diffusion of moisture through the resin rich layer and the debonding between the fibre/matrix interfaces due to the dissolution of the silane coupling agents. Therefore, more developmental work could focus on these aspects to improve further the durability of GFRP bars for civil engineering application. It should be noted that the above conclusions are based on the characterization of a specific bar with a given glass fibre/resin combination and may not directly extend to other types of FRP rebars. Additional tests should be performed on GFRP bars with different types of resin matrixes to provide statistically significant results and conclusions.

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